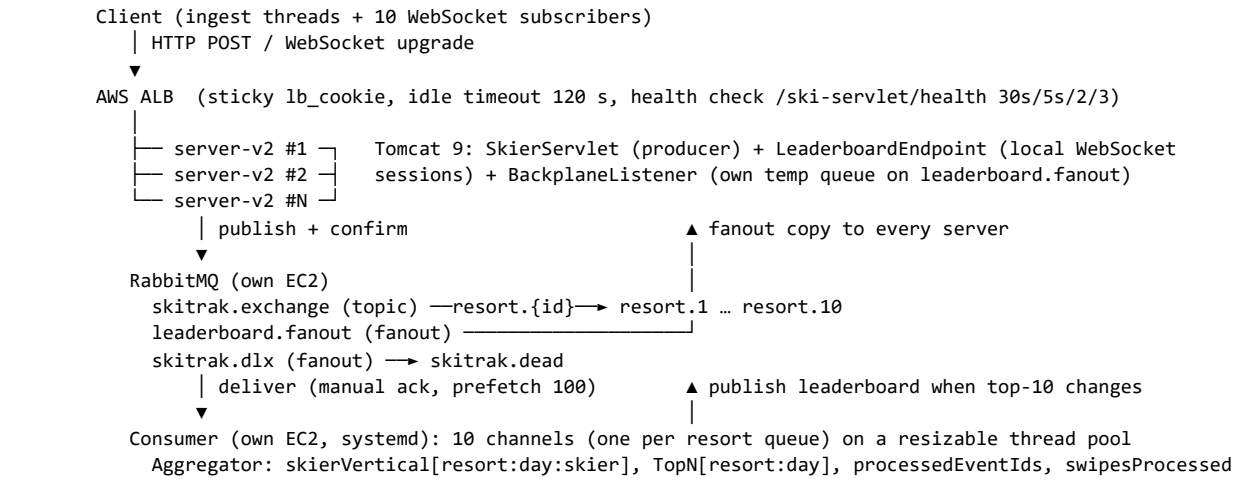
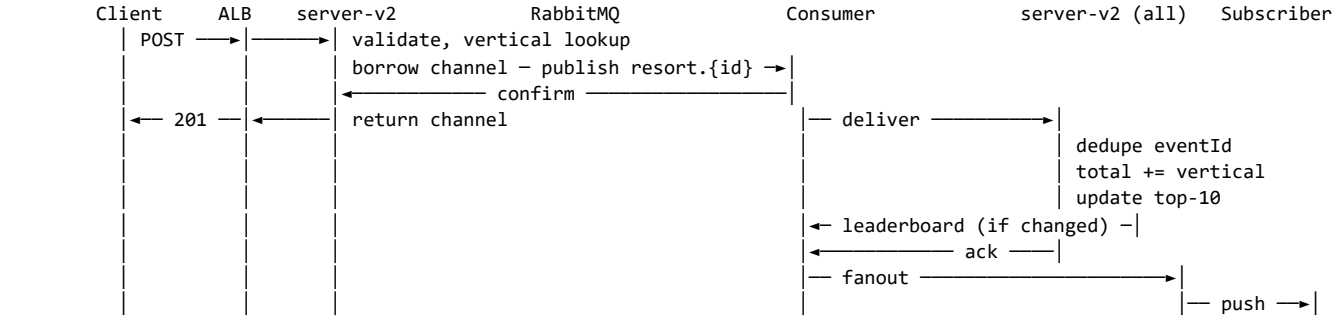
CS6650 Assignment 2 — Juan Yu · Git repository: https://github.com/yulunjuan/cs6650-assignment2

SkiTrak Assignment 2 — Architecture

1. System architecture



2. Swipe → leaderboard sequence



3. Queue topology

Exchange	Type	Binding → queue	Queue arguments
`skitrak.exchange`	topic, durable	`resort.N` → `resort.N` (N = 1..10)	durable; `x-message-ttl` 1 h; `x-max-length` 100,000; `x-overflow` `reject-publish`; `x-dead-letter-exchange` `skitrak.dlx`
`skitrak.dlx`	fanout	→ `skitrak.dead`	durable (expired or rejected swipes)
`leaderboard.fanout`	fanout	→ one server-named, exclusive, auto-delete queue per server-v2	auto-ack

Messages are persistent. The producer uses publisher confirms. `reject-publish` makes a full queue nack the publisher, so the server answers 503 and the client retries. Messages are never dropped silently.

4. Connection locality: broadcast backplane

A subscriber's WebSocket lives on whichever server the ALB picked. The consumer that computes the standings has no connection to it. The consumer publishes each changed leaderboard to `leaderboard.fanout` (routing key = resortID). Every server-v2 binds its own temporary queue, so every server receives every update and pushes it to its local subscribers of that resort.

Why not co-location: partitioning by resort would need one ALB rule and target group per resort. It would pin each resort to one node (no spreading of a hot resort, no failover) and force the consumer into the WebSocket server. **Tradeoff of the backplane:** each update is delivered to all N servers, even those with no subscriber for that resort. Fan-out traffic is updates × N. That is fine for N ≤ 4 and small messages. At larger scale, bind per-resort routing keys only for resorts that have local subscribers.

5. Consumer threading model

- **One channel and one consumer per resort queue.** The RabbitMQ client never runs two deliveries of the same channel at once and keeps them in order, so each resort's swipes are applied in order. Multiple consumers on one queue would reorder a resort's swipes.
- All channels are dispatched on one `ThreadPoolExecutor` (the connection's consumer executor), which serves channels with pending work in turn (fair distribution).
- **Dynamic scaling:** every 5 s, the consumer sums the 10 queue depths. Above 500 it doubles the pool (up to `CONSUMER_MAX_THREADS`), below 50 it halves it (down to `CONSUMER_MIN_THREADS`). Resizing never changes queue ownership, so ordering is unaffected.
- **Consequence:** effective parallelism is capped at 10 (one per resort). Thread counts above 10 are expected to give no further gain. This is the price of per-resort ordering.

6. Delivery guarantees and failure handling

Concern	Mechanism
At-least-once	Manual `basicAck` only after aggregation (and broadcast) succeeded
Idempotency	`processedEventIds` set; checked before and recorded after the state update. The check is race-free because one resort (and hence one eventId) is handled by one thread at a time
Failed processing	3 in-place attempts, then `basicReject(requeue=false)` → `skitrak.dead` (no poison-message loop)
Producer channel safety	`ChannelPool`: one connection, `ArrayBlockingQueue` of 64 confirm-mode channels; borrow → publish → `waitForConfirmsOrDie` → return
Producer reconnection	On borrow, a closed channel is replaced; a closed connection is re-opened (synchronized, 1 s fail-fast backoff while the broker is down)
Consumer / backplane reconnection	Client automatic recovery (connection, channels, consumers, server-named queue + binding); redeliveries are deduped
Consumer health	`GET :8081/health` (503 when disconnected); systemd `Restart=always`; cron health check restarts a hung process
Graceful shutdown	Consumer shutdown hook closes the connection (unacked messages requeue); `ServletContextListener` closes pool and backplane
Server health	ALB `/ski-servlet/health`; the webapp fails to deploy without a broker, so the target turns unhealthy

7. Load-balancing configuration

ALB → one target group of server-v2 instances on port 8080. Stickiness `lb_cookie`, 1 day. Idle timeout 120 s (> 60 s, so idle WebSockets survive). Health check `/ski-servlet/health`, interval 30 s, timeout 5 s, healthy 2, unhealthy 3. WebSocket upgrades pass through the HTTP listener natively. Once upgraded, a WebSocket stays on the target that accepted it. Because every server receives every leaderboard update (section 4), a subscriber is served correctly no matter which target it lands on.

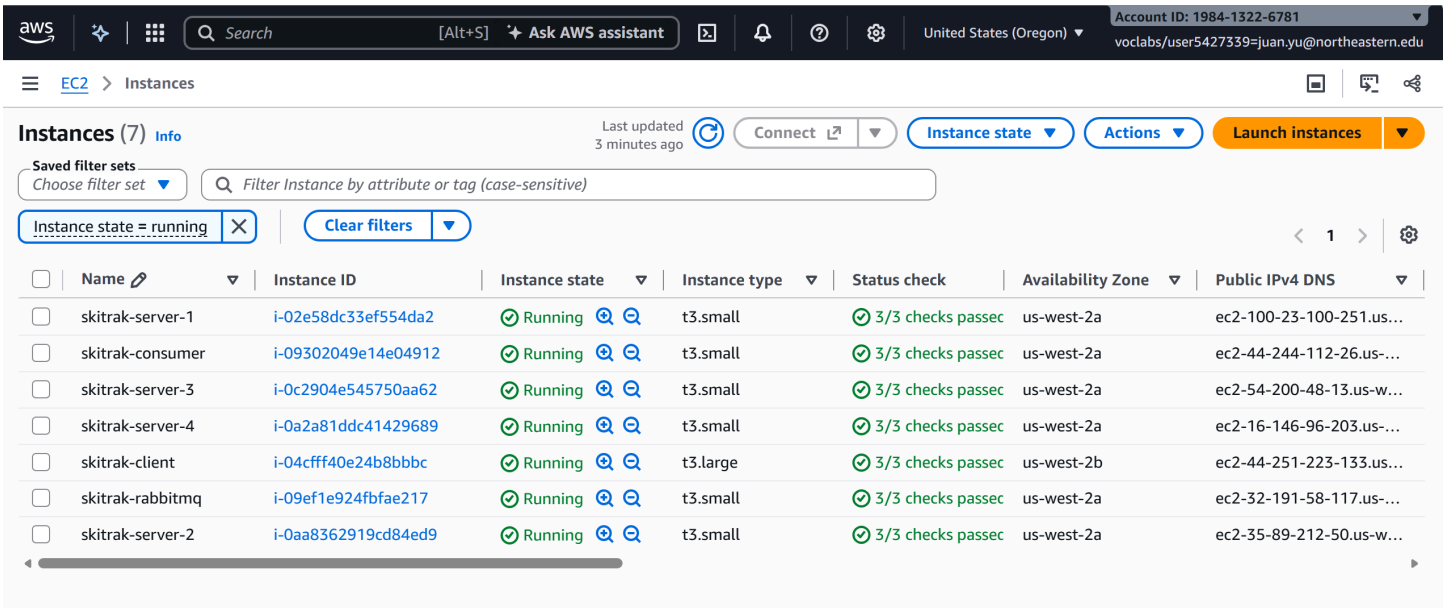
SkiTrak Assignment 2 — Test Results

1. Test environment

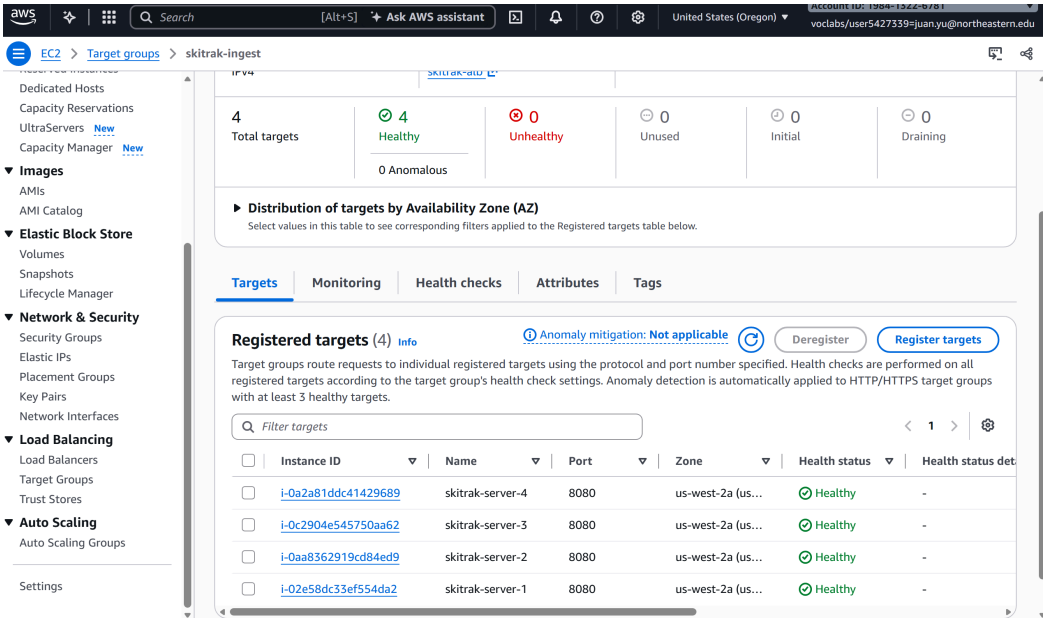
All instances are in AWS us-west-2. Times are UTC unless noted; the RabbitMQ UI shows UTC−6.

Role	Name	Type	AZ	Notes
Broker	skitrak-rabbitmq	t3.small	us-west-2a	Amazon Linux 2023, Docker `rabbitmq:3.13-management` (RabbitMQ 3.13.7, Erlang 26.2.5.16)
Consumer	skitrak-consumer	t3.small	us-west-2a	Amazon Linux 2023, Corretto 17, systemd, `-Xmx1200m`
Ingest servers	skitrak-server-1..4	t3.small	us-west-2a	Launched from an AMI of my A1 server (Tomcat 9)
Load client	skitrak-client	t3.large	us-west-2b	Same region: RTT ~8–12 ms instead of ~142 ms from my laptop
Load balancer	skitrak-alb	ALB, internet-facing	2a + 2b	HTTP:80 → target group skitrak-ingest (HTTP:8080)

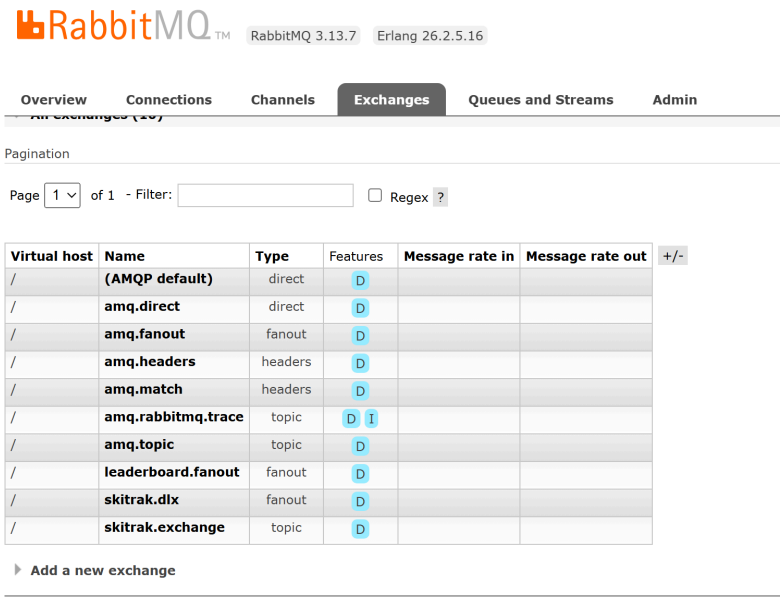
- Target group health check: `/ski-servlet/health`, interval 30 s, timeout 5 s, healthy 2, unhealthy 3. Stickiness: load-balancer-generated cookie, 1 day. ALB idle timeout: 120 s. `curl -i` through the ALB returned `Set-Cookie: AWSALB=...` and `AWSALBCORS=...`.
- Topology, as confirmed in the RabbitMQ UI:
  - exchanges `skitrak.exchange` (topic), `leaderboard.fanout` (fanout), `skitrak.dlx` (fanout)
  - queues `resort.1..10` (D, TTL, Lim, Ovfl, DLX), `skitrak.dead` (D), and one `amq.gen-*` queue per server (AD, Excl)
- Startup checks: each server logged `server-v2 <host> connected to RabbitMQ, channel pool size 64`. The consumer health check returned `{"status":"UP","swipesProcessed":0,"queueDepth":0}`.
- Procedure: before each run I restarted the consumer (`systemctl restart skitrak-consumer`) to clear the in-memory aggregates and the eventId set. The exception is alb4-t512 (see §7).



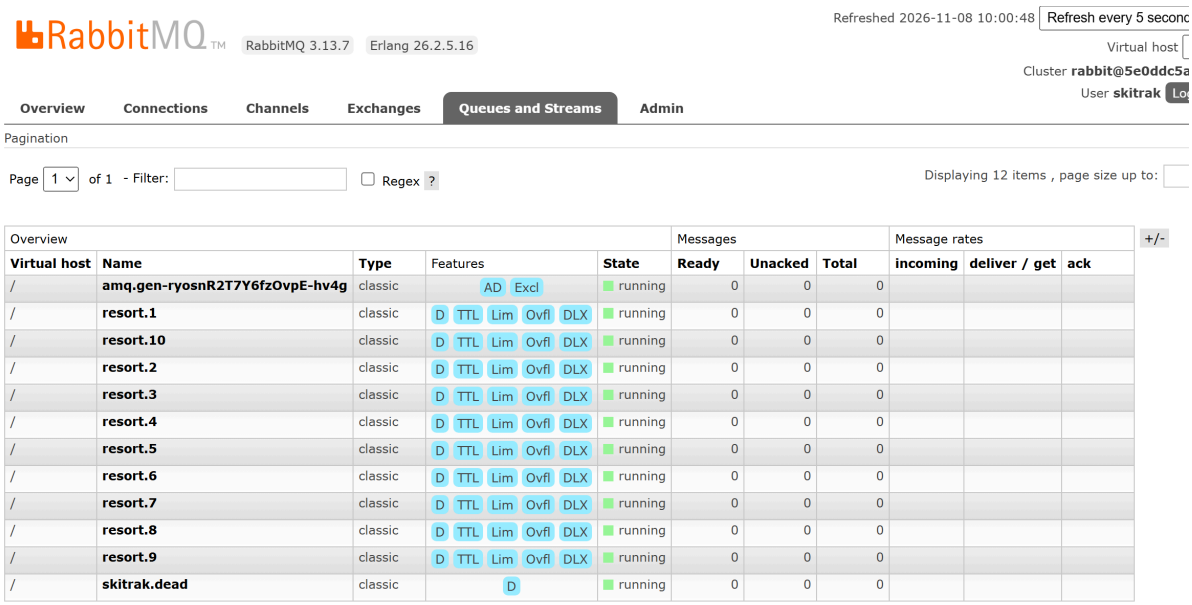
All SkiTrak instances running: 4 ingest servers, consumer and broker (t3.small), and the client (t3.large).



Target group skitrak-ingest: 4 registered targets (skitrak-server-1..4, port 8080), all Healthy.



RabbitMQ exchanges: `skitrak.exchange` (topic), `leaderboard.fanout` (fanout) and `skitrak.dlx` (fanout), all durable, next to the built-in exchanges.



RabbitMQ queues (taken 10:00 local, with one server deployed): `resort.1..10` with D/TTL/Lim/Ovfl/DLX, `skitrak.dead` (D), and that server's `amq.gen-*` backplane queue (AD, Excl).

2. Run results

Every run had 0 failed swipes, 0 retries, 0 connection errors and 0 WebSocket reconnections. In every run, each resort received about 10% of the swipes.

Label	Servers	Client threads	Swipes	Wall time (ms)	Single-thread RTT	Warmup (32 thr)	Main phase	Overall
smoke (laptop → server-1)	1	64	50,000	317,468	142.37 ms	142/s	194/s	157/s
single-500k	1	256	500,000	101,123	9.40 ms	2,627/s	5,262/s	**4,944/s**
alb2-500k	2	256	500,000	85,791	9.68 ms	2,828/s	6,284/s	**5,828/s**
alb4-500k	4	256	500,000	86,152	10.54 ms	2,827/s	6,254/s	**5,804/s**
alb4-1m	4	256	1,000,000	161,924	11.53 ms	2,795/s	6,433/s	**6,176/s**
alb4-t512 (consumer not restarted)	4	512	500,000	82,608	8.97 ms	2,940/s	6,525/s	6,053/s
alb4-t512b (clean)	4	512	500,000	84,770	8.26 ms	2,641/s	6,442/s	5,898/s

Label	Ingest latency mean / median / p95 / p99 / max (ms)	Push latency median / p95 / p99 (ms)	Leaderboard updates	Little's Law prediction (main phase)
smoke	259.81 / 250 / 368 / 475 / 1626	236 / 378 / 1696	640	450/s (warmup 225/s)
single-500k	45.34 / 25 / 137 / 220 / 2234	4 / 32 / 118	1,353	27,234/s
alb2-500k	38.47 / 37 / 66 / 99 / 455	9 / 21 / 56	1,350	26,457/s
alb4-500k	38.69 / 37 / 63 / 87 / 251	12 / 25 / 70	1,300	24,281/s
alb4-1m	38.67 / 37 / 62 / 91 / 374	14 / 25 / 56	1,636	22,203/s
alb4-t512	73.23 / 73 / 128 / 179 / 719	18 / 42 / 339,863	298	57,058/s
alb4-t512b	74.27 / 73 / 130 / 178 / 965	11 / 28 / 71	1,319	62,002/s

Full outputs and CSVs are in `results/<label>-output.txt`, `-latency.csv` and `-throughput.csv`.

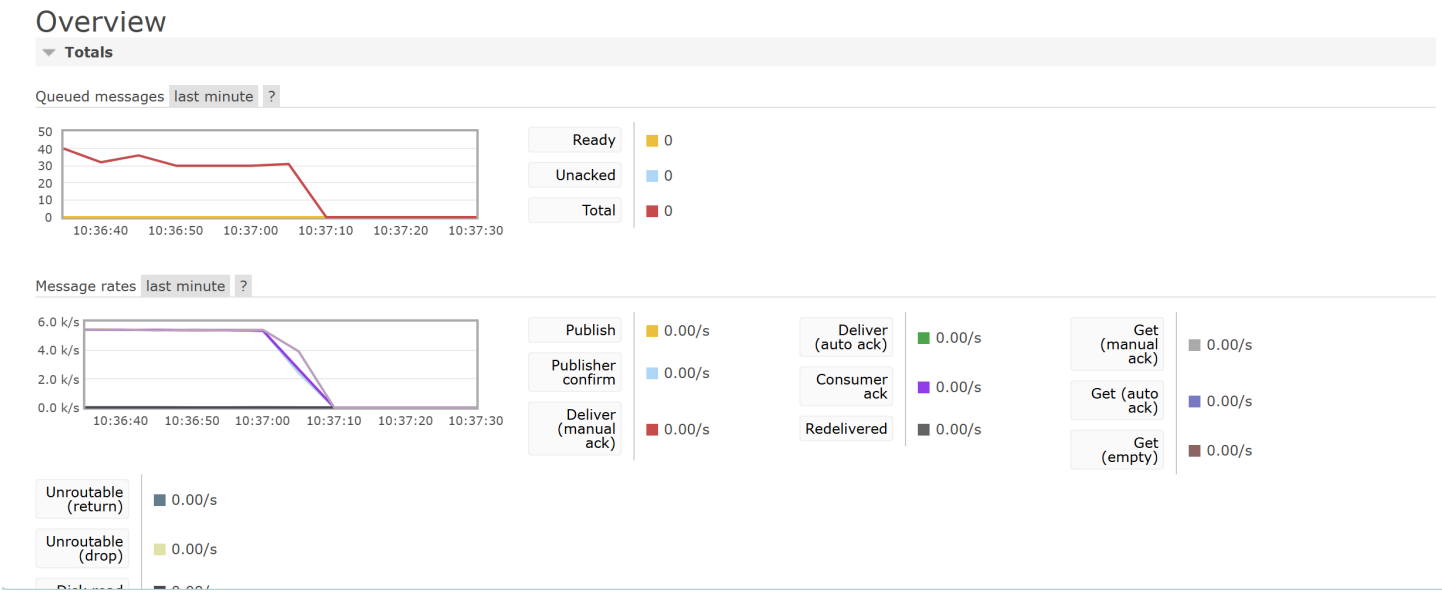
3. Real leaderboard evidence

Subscribers receive real, swipe-driven standings in every run. Each “Last Leaderboard Received” section lists real skierIDs with accumulated vertical: about 600–980 ft in the smoke run, ~1,250–1,800 ft in the 500K runs, and ~1,670–2,120 ft in the 1M run. Here is the full single-500k block:


```
===== Last Leaderboard Received =====
Resort  1: {"resortID":"1","dayID":"1","serverTimestamp":"2026-11-08T16:37:02.835957402Z","top":
[{"rank":1,"skierID":"94506","vertical":1434},{"rank":2,"skierID":"10517","vertical":1391},
{"rank":3,"skierID":"29384","vertical":1374},{"rank":4,"skierID":"84823","vertical":1363},
{"rank":5,"skierID":"5746","vertical":1327},{"rank":6,"skierID":"14014","vertical":1321},
{"rank":7,"skierID":"15682","vertical":1320},{"rank":8,"skierID":"49864","vertical":1304},
{"rank":9,"skierID":"12964","vertical":1294},{"rank":10,"skierID":"95723","vertical":1294}]}
Resort  2: {"resortID":"2","dayID":"1","serverTimestamp":"2026-11-08T16:37:00.474954282Z","top":
[{"rank":1,"skierID":"23265","vertical":1572},{"rank":2,"skierID":"49680","vertical":1469},
{"rank":3,"skierID":"15576","vertical":1443},{"rank":4,"skierID":"40351","vertical":1442},
{"rank":5,"skierID":"12788","vertical":1356},{"rank":6,"skierID":"59908","vertical":1356},
{"rank":7,"skierID":"25336","vertical":1312},{"rank":8,"skierID":"77835","vertical":1304},
{"rank":9,"skierID":"67544","vertical":1267},{"rank":10,"skierID":"17233","vertical":1260}]}
Resort  3: {"resortID":"3","dayID":"1","serverTimestamp":"2026-11-08T16:37:03.470546358Z","top":
[{"rank":1,"skierID":"75385","vertical":1529},{"rank":2,"skierID":"24542","vertical":1512},
{"rank":3,"skierID":"44151","vertical":1486},{"rank":4,"skierID":"23481","vertical":1372},
{"rank":5,"skierID":"4388","vertical":1347},{"rank":6,"skierID":"3858","vertical":1320},
{"rank":7,"skierID":"23941","vertical":1320},{"rank":8,"skierID":"37516","vertical":1287},
{"rank":9,"skierID":"77623","vertical":1260},{"rank":10,"skierID":"81159","vertical":1242}]}
Resort  4: {"resortID":"4","dayID":"1","serverTimestamp":"2026-11-08T16:36:59.198375176Z","top":
[{"rank":1,"skierID":"25912","vertical":1434},{"rank":2,"skierID":"69311","vertical":1433},
{"rank":3,"skierID":"28838","vertical":1415},{"rank":4,"skierID":"19844","vertical":1407},
{"rank":5,"skierID":"45233","vertical":1346},{"rank":6,"skierID":"31583","vertical":1336},
{"rank":7,"skierID":"5237","vertical":1330},{"rank":8,"skierID":"34281","vertical":1322},
{"rank":9,"skierID":"75999","vertical":1321},{"rank":10,"skierID":"52251","vertical":1312}]}
Resort  5: {"resortID":"5","dayID":"1","serverTimestamp":"2026-11-08T16:37:03.280388499Z","top":
[{"rank":1,"skierID":"48660","vertical":1582},{"rank":2,"skierID":"94554","vertical":1581},
{"rank":3,"skierID":"47813","vertical":1511},{"rank":4,"skierID":"78545","vertical":1468},
{"rank":5,"skierID":"86687","vertical":1451},{"rank":6,"skierID":"44483","vertical":1443},
{"rank":7,"skierID":"50314","vertical":1390},{"rank":8,"skierID":"38921","vertical":1365},
{"rank":9,"skierID":"12481","vertical":1355},{"rank":10,"skierID":"49618","vertical":1354}]}
Resort  6: {"resortID":"6","dayID":"1","serverTimestamp":"2026-11-08T16:36:58.058541977Z","top":
[{"rank":1,"skierID":"66864","vertical":1625},{"rank":2,"skierID":"20136","vertical":1580},
{"rank":3,"skierID":"70180","vertical":1469},{"rank":4,"skierID":"37695","vertical":1407},
{"rank":5,"skierID":"14095","vertical":1400},{"rank":6,"skierID":"92131","vertical":1381},
{"rank":7,"skierID":"6747","vertical":1312},{"rank":8,"skierID":"87871","vertical":1312},
{"rank":9,"skierID":"5923","vertical":1286},{"rank":10,"skierID":"88359","vertical":1269}]}
Resort  7: {"resortID":"7","dayID":"1","serverTimestamp":"2026-11-08T16:37:03.515046298Z","top":
[{"rank":1,"skierID":"29470","vertical":1573},{"rank":2,"skierID":"49728","vertical":1425},
{"rank":3,"skierID":"25597","vertical":1380},{"rank":4,"skierID":"16426","vertical":1373},
{"rank":5,"skierID":"17521","vertical":1373},{"rank":6,"skierID":"88500","vertical":1347},
{"rank":7,"skierID":"12058","vertical":1311},{"rank":8,"skierID":"21824","vertical":1303},
{"rank":9,"skierID":"64776","vertical":1295},{"rank":10,"skierID":"67741","vertical":1293}]}
Resort  8: {"resortID":"8","dayID":"1","serverTimestamp":"2026-11-08T16:37:00.778227809Z","top":
[{"rank":1,"skierID":"32420","vertical":1529},{"rank":2,"skierID":"86398","vertical":1477},
{"rank":3,"skierID":"40751","vertical":1469},{"rank":4,"skierID":"83062","vertical":1433},
{"rank":5,"skierID":"98197","vertical":1426},{"rank":6,"skierID":"10582","vertical":1417},
{"rank":7,"skierID":"46528","vertical":1356},{"rank":8,"skierID":"39711","vertical":1348},
{"rank":9,"skierID":"32188","vertical":1348},{"rank":10,"skierID":"18178","vertical":1329}]}
Resort  9: {"resortID":"9","dayID":"1","serverTimestamp":"2026-11-08T16:37:02.982358472Z","top":
[{"rank":1,"skierID":"15612","vertical":1765},{"rank":2,"skierID":"61336","vertical":1505},
{"rank":3,"skierID":"22830","vertical":1383},{"rank":4,"skierID":"99354","vertical":1381},
{"rank":5,"skierID":"25727","vertical":1380},{"rank":6,"skierID":"15007","vertical":1330},
{"rank":7,"skierID":"83525","vertical":1313},{"rank":8,"skierID":"43774","vertical":1313},
{"rank":9,"skierID":"8083","vertical":1311},{"rank":10,"skierID":"59795","vertical":1304}]}
Resort 10: {"resortID":"10","dayID":"1","serverTimestamp":"2026-11-08T16:37:00.058022458Z","top":
[{"rank":1,"skierID":"54955","vertical":1504},{"rank":2,"skierID":"83188","vertical":1469},
{"rank":3,"skierID":"40311","vertical":1460},{"rank":4,"skierID":"71138","vertical":1459},
{"rank":5,"skierID":"54870","vertical":1390},{"rank":6,"skierID":"45912","vertical":1372},
{"rank":7,"skierID":"90771","vertical":1338},{"rank":8,"skierID":"34351","vertical":1330},
{"rank":9,"skierID":"75826","vertical":1329},{"rank":10,"skierID":"12542","vertical":1322}]}
```


4. Queue metrics

single-500k (RabbitMQ Overview, last minute): publish, deliver and consumer-ack rates overlapped at approx. 5.4k msg/s. Queued messages stayed at approx. 30–40 during the run and returned to 0 afterwards.

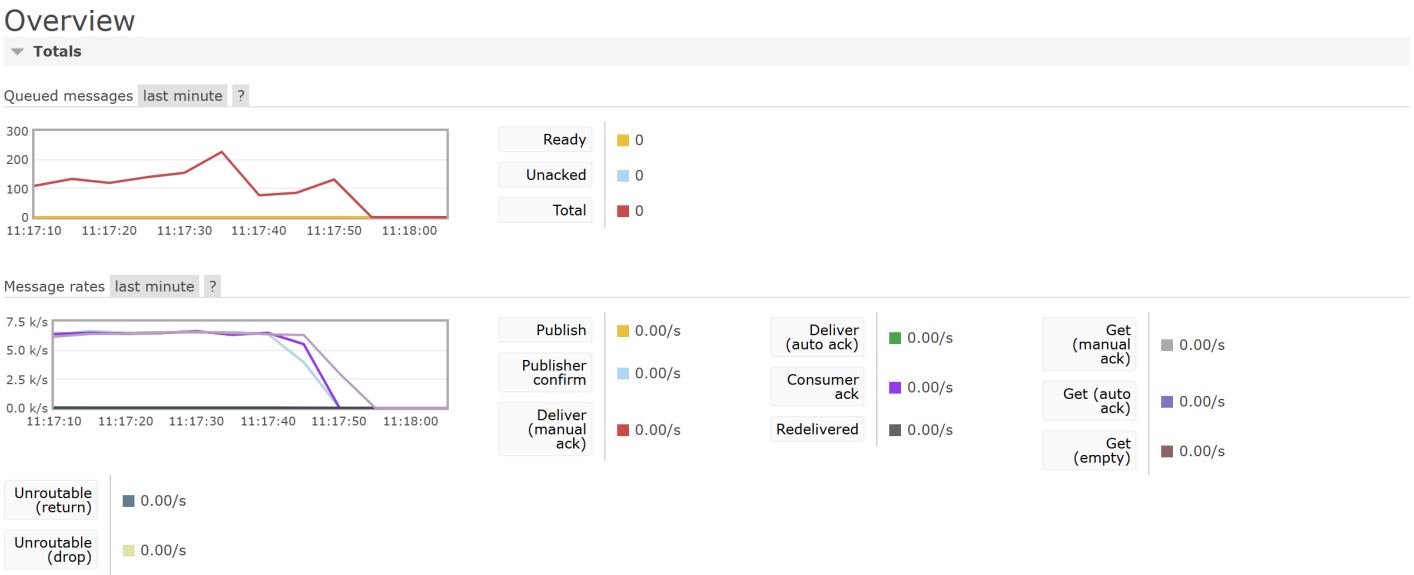


RabbitMQ Overview, single-500k run: rates overlap at ~5.4k msg/s and the queue holds a low plateau.

alb4-1m (`queue_monitor.py` every 5 s, summarized by `summarize_queue.py`):

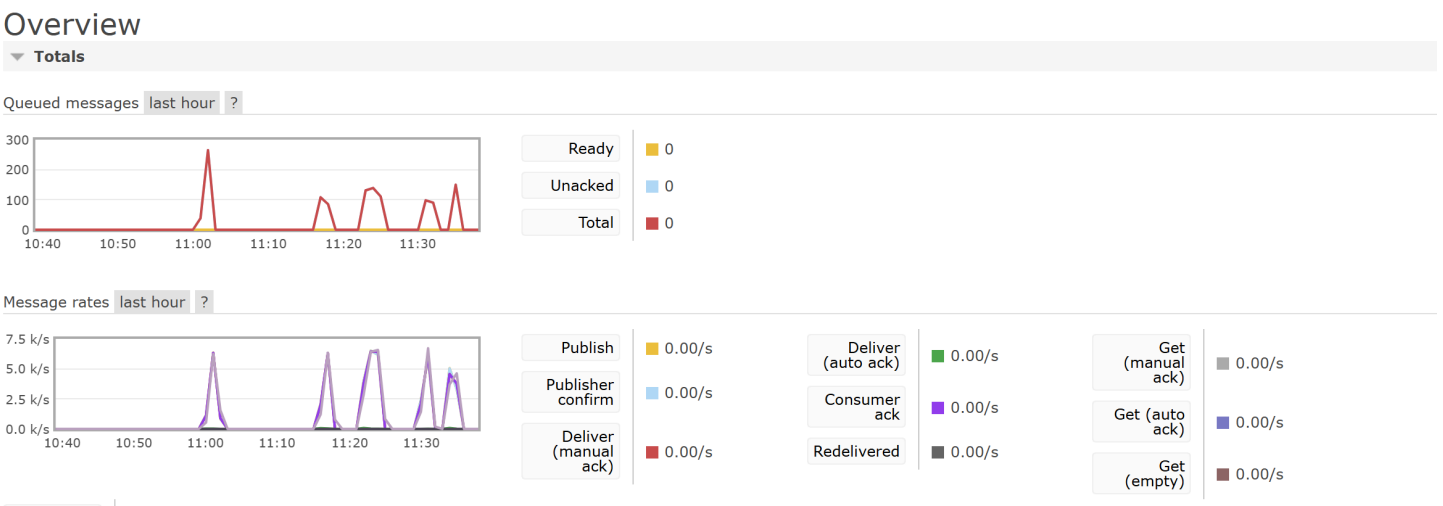
```
Samples (active / total): 31 / 45
Peak queue depth:   265
Avg queue depth:    105
Avg producer rate:  5932 msg/s
Avg consumer rate:  6047 msg/s
Peak producer rate: 7303 msg/s
Peak consumer rate: 6840 msg/s
```

Depth fluctuated between roughly 60 and 180 with no upward trend, which is a stable plateau.

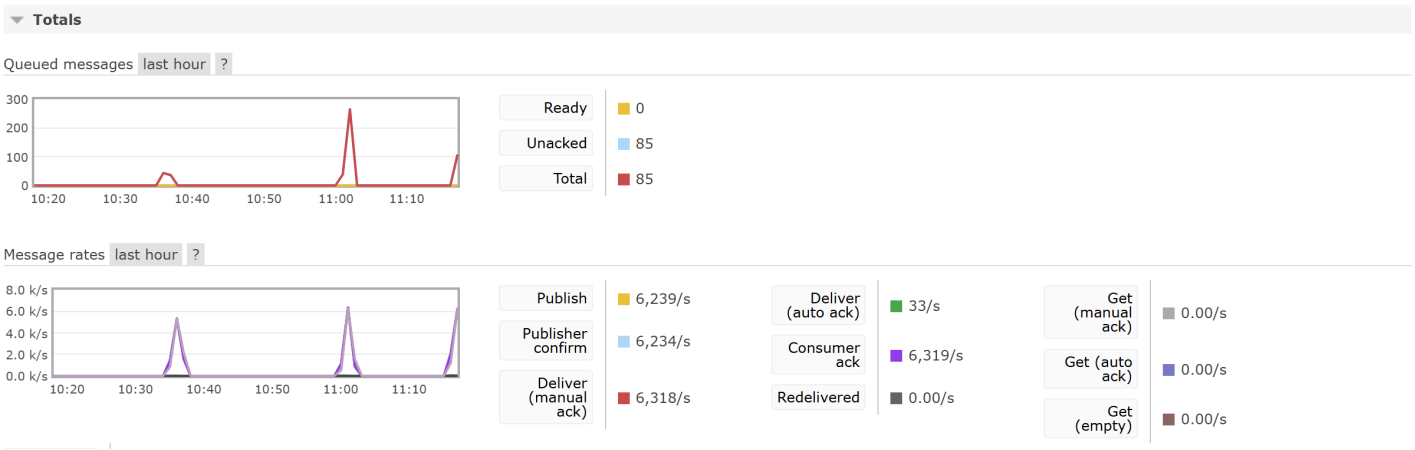


RabbitMQ overview, last minute, end of the alb4-500k run (4 instances, 256 client threads).

All load-balanced runs (Overview, last hour, local UTC−6): peaks at 11:01 alb2 (queue approx. 260, the highest of all runs), 11:16 alb4-500k (approx. 100), 11:22 alb4-1m (the widest), 11:29 alb4-t512 and 11:34 alb4-t512b (approx. 150). The queue returned to 0 after every run. At the one-hour scale each point averages a time window, so plateaus look like spikes and rates look lower than the per-run measurements.



RabbitMQ Overview, last hour (local UTC−6). Queue peaks: 11:01 alb2-500k, 11:16 alb4-500k, 11:22 alb4-1m, 11:29 alb4-t512, 11:34 alb4-t512b. Queued messages return to 0 after each run.



RabbitMQ overview, last hour, captured at 11:18 (UTC−6). It shows the single-instance run (10:36, queue ~40) and the 2-instance run (11:01, queue ~260). The right edge is the alb4-500k run still in progress.

All runs stayed far below the 1,000-message depth target.

5. Load distribution and stickiness

- **2-server run.** The Tomcat access log on server-2 counted **250,049** `POST /ski-servlet/skiers` requests, which is 50.01% of 500,000. Server-1 therefore handled 249,951. Only the 2026-11-08 file is from this test; the older files came from the A1 AMI.

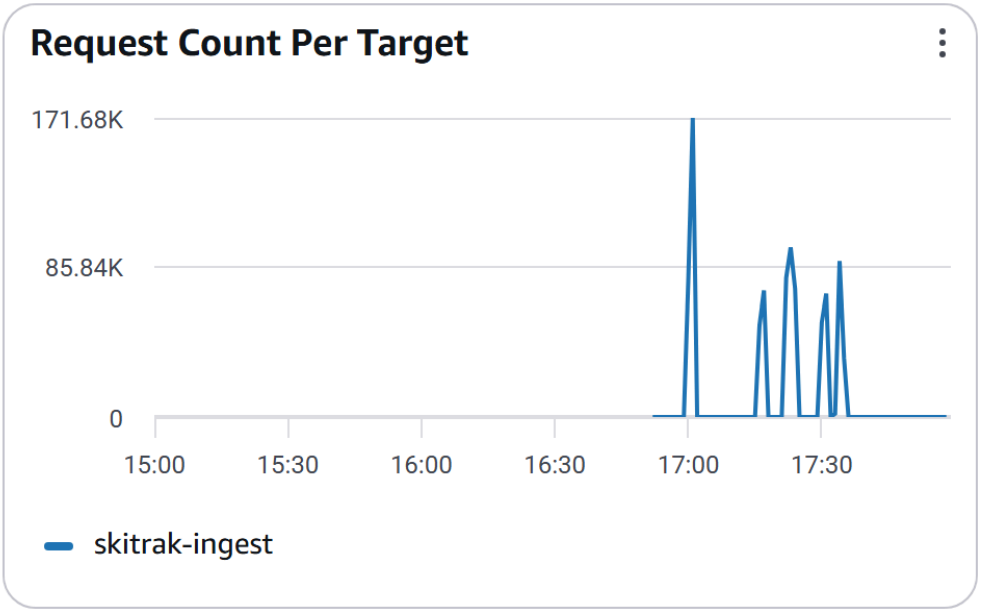
```
$ grep -c "POST /ski-servlet/skiers" /opt/tomcat9/logs/localhost_access_log.*
/opt/tomcat9/logs/localhost_access_log.2026-09-12.txt:0
/opt/tomcat9/logs/localhost_access_log.2026-10-16.txt:2046679
/opt/tomcat9/logs/localhost_access_log.2026-11-08.txt:250049
```

- **Sticky cookie.** `curl -i http://<ALB>/ski-servlet/health`:

```
HTTP/1.1 200 OK
Date: Sun, 08 Nov 2026 17:00:00 GMT
Content-Type: application/json;charset=UTF-8
Set-Cookie: AWSALB=...; Expires=Sun, 15 Nov 2026 17:00:00 GMT; Path=/
Set-Cookie: AWSALBCORS=...; Expires=Sun, 15 Nov 2026 17:00:00 GMT; Path=/;SameSite=None
Content-Length: 16

{"message":"OK"}
```

- **CloudWatch RequestCountPerTarget** shows the traffic spike at ~17:00 UTC. This metric is the average per healthy target, not a per-instance breakdown.



CloudWatch RequestCountPerTarget, target group skitrak-ingest (UTC). The tallest spike is at ~17:00 (alb2-500k, 2 targets); the smaller spikes after 17:15 are the 4-target runs. The metric is the average per healthy target.

- **Stickiness.** The ingest HTTP client does not store cookies, so ALB stickiness did not pin POSTs to one server, and they were spread evenly. Each WebSocket stayed on the server that accepted it: there were 0 reconnections in every run. Because of the backplane, every server receives every leaderboard update, so a subscriber is served correctly on any target.

6. Resource utilization

vmstat on the RabbitMQ host (5 s interval, run alb4-t512; consumer state does not affect ingest load):

	us	sy	id	wa	bo
main phase	65–78	20–25	1–2	0 (7–8 only in the first seconds)	~13,600–14,900 KB/s (persistent writes)
alb4-t512b (only the last ~10 s captured)	72–75	23–27	1–2		

So the broker CPU was 95–99% busy in the main phase. Main-phase excerpt:

```
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 8  1      0 338764   2168 1151564    0    0     0 24822 20325 37971 58 29  6  8  0
 4  0      0 318252   2168 1151252    0    0     0 14946 18306 13015 72 23  5  0  0
 2  0      0 332280   2168 1131560    0    0     0 14468 17413 11587 76 22  2  0  0
 2  1      0 328744   2168 1126808    0    0     0 13946 17309 11240 78 21  1  0  0
 6  0      0 325148   2168 1143332    0    0     0 14246 17236 11280 76 23  1  0  0
 2  1      0 312456   2168 1159196    0    0     0 13832 18007 10943 76 23  1  0  0
 4  0      0 316800   2168 1139636    0    0     0 14307 18659 11773 78 21  1  0  0
 2  0      0 305996   2168 1156052    0    0     0 13953 17458 10949 77 22  1  0  0
```

vmstat on the client (alb4-t512): idle was 0–6% during warmup/ramp-up. In the main phase, idle was 18–28%, steal 5–8% and the run queue reached up to ~49. Main-phase excerpt:

```
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
49  0      0 5871884   2168 764516    0    0     0     0 21606 50089 56 36  7  0  2
10  0      0 5863824   2168 764516    0    0     0     0 19189 41150 39 28 28  0  6
16  0      0 5861332   2168 764516    0    0     0    23 22502 49478 41 33 20  0  5
18  0      0 5859892   2168 764516    0    0     0     0 21771 49081 41 33 19  0  7
16  0      0 5857860   2168 764516    0    0     0     0 21307 48637 43 34 18  0  6
19  0      0 5854360   2168 764516    0    0     0     0 21625 48948 39 32 21  0  8
23  0      0 5853192   2168 764516    0    0     0     0 21521 48497 38 32 22  0  8
```

EC2 CPU utilization, RabbitMQ host (CloudWatch 5-minute averages, approx.):

Time	Run	CPU
17:00	alb2	~26%
17:20	1M	43.53% (peak 5-minute average)

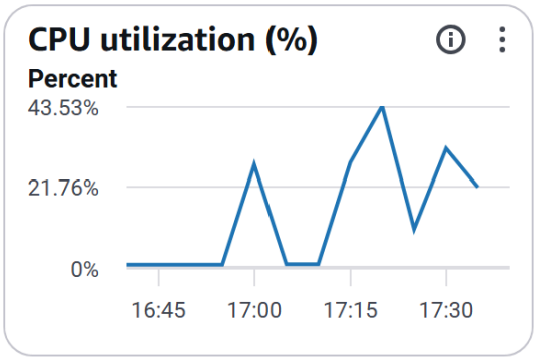
Time	Run	CPU
17:30	t512	~32%

The CPU credit balance kept rising, so the instance was not throttled. Each run lasted only 1.5–2.7 min, so the 5-minute averages include idle time and understate in-run load. vmstat is the direct measurement.

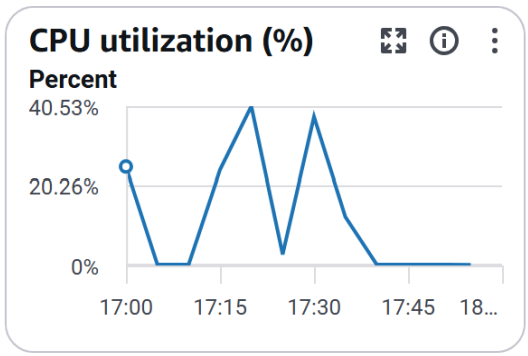
EC2 CPU utilization, all roles (CloudWatch 5-minute averages, UTC). The top y-axis label of each chart is the highest 5-minute average in the window, not an instantaneous value during a run:

Instance	Peak 5-minute average	What the chart shows
skitrak-rabbitmq	43.53%	Highest of the four. Rises at ~17:00, peaks at ~17:20 (1M run), stays elevated through ~17:35
skitrak-client	40.53%	Peaks around 17:20 with a second peak near 17:30, then drops to 0 after ~17:40
skitrak-server-1	14.31%	The same two-peak shape at a much lower level
skitrak-consumer	12.94%	The same two-peak shape at a much lower level

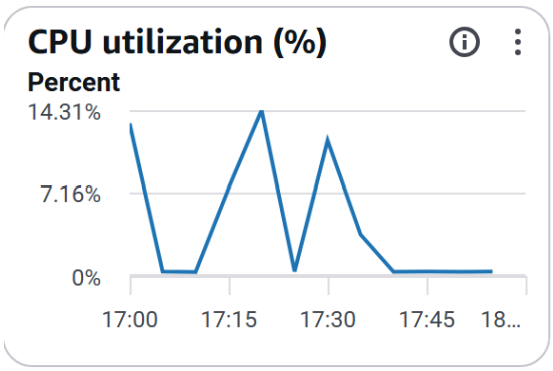
The ingest server and the consumer stay low, while the broker carries the highest load, which matches the vmstat result.



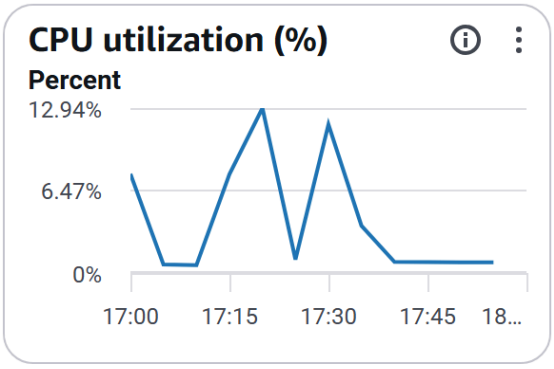
skitrak-rabbitmq CPU, 5-minute averages (peak 5-minute average 43.53%).



skitrak-client CPU, 5-minute averages (peak 5-minute average 40.53%).



skitrak-server-1 CPU, 5-minute averages (peak 5-minute average 14.31%).



skitrak-consumer CPU, 5-minute averages (peak 5-minute average 12.94%).

7. Analysis

Laptop smoke test. From my laptop, single-thread RTT was 142 ms and the client measured 157 swipes/s overall. Little's Law predicted 225 req/s for the 32-thread warmup. Under load, though, median latency rose to ~250 ms, and 32 / 0.25 s ≈ 128 req/s matches the measured 142/s. The consumer was idle and queue depth stayed at 0, so the bottleneck was the network from my laptop. For this reason, all performance runs used a client EC2 instance in the same region.

Single instance. The run processed 500,000 swipes with no failures at 4,944 swipes/s overall (5,262/s main phase). Publish, deliver and ack rates stayed equal at ~5.4k msg/s, and queue depth stayed at ~30–40, a stable plateau, so the consumer kept up. Little's Law predicted 27,234 req/s from the 9.4 ms single-thread RTT. Under load, mean latency rose to ~45 ms, and 256 / 0.045 s ≈ 5,700 req/s matches the measurement.

1 → 2 instances. Throughput rose 18% (4,944 → 5,828 overall; 5,262 → 6,284 main phase) and tail latency roughly halved (p95 137 → 66 ms, p99 220 → 99 ms). Spreading requests over two servers removed most of the queueing on a single server. The ALB split the load evenly (250,049 vs 249,951).

2 → 4 instances. Throughput did not increase (main phase 6,284 → 6,254/s), and mean latency stayed at ~38 ms. If the ingest servers were still the bottleneck, halving each server's load would have lowered latency. The bottleneck had therefore moved downstream of the servers.

1M stress test. The run processed 1,000,000 swipes with no failures at 6,176 swipes/s overall (6,433/s main phase), the same rate as the 500K run, with p99 at 91 ms. Over ~2.5 min, queue depth stayed between ~60 and 180 (peak 265, average 105) with no upward trend. The average consumer rate (6,047 msg/s) matched the producer rate (5,932 msg/s). The system was stable under sustained load, and the consumer was not the bottleneck.

256 → 512 client threads. Main-phase throughput rose only 3% (6,254 → 6,442/s), while mean latency nearly doubled (39 → 74 ms). The extra concurrency only added queueing, so the system was saturated downstream of the client. vmstat showed the RabbitMQ host at 95–99% CPU with near-zero I/O wait, while the client still had ~20% idle CPU in the main phase.

Bottleneck. The bottleneck is the CPU of the single t3.small RabbitMQ broker. Each swipe costs the broker a persistent publish, a publisher confirm, a delivery and a manual ack, plus leaderboard fan-out to every server. Adding ingest servers beyond two therefore cannot raise throughput. Options:

- a larger broker instance
- batched or asynchronous publisher confirms
- batched consumer acks
- throttled leaderboard broadcasts
- non-persistent messages, if some loss were acceptable

Measurement finding (push latency). In run alb4-t512 the consumer was not restarted, because the restart command failed on a pasted control sequence. Push-latency p99 jumped to ~340 s, about the gap since the previous run. The server sends its cached latest leaderboard to each new subscriber on connect. That snapshot came from the previous run, so its serverTimestamp was minutes old. Normally these 10 initial messages are diluted by more than 1,000 live updates. Here, because the top 10 already held high values from the previous run, only 298 updates occurred, and the stale snapshots dominated p99. After restarting the consumer (alb4-t512b), p99 returned to 71 ms. A more accurate push-latency measurement should exclude each subscriber's first message.

Known limitation (eventId dedupe). The server generates the eventId. Dedupe therefore catches broker redeliveries, but not a client retry after a lost 201, because the retry gets a new eventId. Fixing this needs a client-generated ID, which changes the A1 API.

8. Changes made during deployment

- `deployment/consumer/skitrak-consumer.service`: `-Xmx2g` → `-Xmx1200m`. The consumer host (t3.small) has 2 GB of total RAM, so a 2 GB heap risked an OOM kill.
- The client runs on a separate EC2 instance. The `results/` folder must be created on that instance before `tee`.
- The consumer was restarted before each run so that aggregates and the dedupe set do not carry over between runs.

9. Not completed

- Consumer thread sweep (10/20/40/80): not run. By design, one consumer per resort queue caps effective parallelism at 10, and the measured bottleneck is the broker, so more consumer threads are not expected to help.
- Client thread sweep: only 256 and 512 were run. 64 and 128 were not run on EC2; 64 was used only in the laptop smoke test.
- Failover test (stopping a server mid-run) and broker-restart reconnection test: not run.
- Batch-ack and async-confirm tuning: not tried.